1. Competency #3: Implement a given strategy to determine a mathematical solution

(Highly relevant)

- The problem provides all necessary data and implies a standard strategy: use the formula $t = d/v$.
- Students must recognize that average speed over a whole trip allows direct calculation of total time.
- This is a textbook example of applying a known mathematical procedure to reach a solution.

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2. Competency #7: Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

(Highly relevant)

- The core task involves numerical manipulation: dividing 6 by 18 and converting hours to minutes.
- While no graphs or algebra are used, the operation on real-world quantitative data is central.

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3. Competency #1: Perform a simple calculation

(Moderately relevant)

- The arithmetic (6 ÷ 18 = 1/3; 1/3 h = 20 min) is straightforward but essential.
- As with qs1, this competency supports higher-level actions but is not the primary focus.

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4. Competency #13: Use an understanding of the context to guide or expedite the mathematical solving process

(Mildly relevant)

- The context (a bike trip with a speedometer showing average speed) signals that the average speed applies to the *entire* journey, justifying the use of the total-distance/total-time formula.
- Misinterpreting "average speed" as possibly varying could lead a student to choose D ("not possible to tell"), so contextual understanding helps avoid that trap.
- Thus, real-world knowledge about what a speedometer reports aids correct modeling.

5. Competency #8: Articulate a solution, showing and/or summarising and presenting intermediate mathematical results

(Weakly relevant in current format)

- Since this is a multiple-choice question, articulation isn't required.
- However, if used in a classroom setting where justification is expected, explaining why D is incorrect (i.e., average speed *does* allow time calculation) would engage this competency.

1. 能力 #3：实施给定策略以确定数学解

（高度相关）

- 题目提供了完整数据，并隐含了一个标准策略：使用公式 $t = d/v$ 。
- 学生需理解“全程平均速度”足以用于计算总时间。
- 这是应用已知数学程序求解的典型范例。

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2. 能力 #7：操作数字、图表与统计信息、代数表达式与方程、几何表示等

（高度相关）

- 核心任务涉及数值运算：6 ÷ 18，以及将小时换算为分钟。
- 虽然未使用图表或代数，但对现实情境中数量信息的操作是关键。

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3. 能力 #1：执行简单计算

（中等相关）

- 计算过程（6 ÷ 18 = 1/3；1/3 小时 = 20 分钟）简单但必不可少。
- 与 qs1 类似，此能力为基础支撑，而非主要考查目标。

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4. 能力 #13：利用对情境的理解来指导或加速数学解题过程

（弱至中等相关）

- 情境（骑车、速度表显示“全程平均速度”）暗示该平均速度适用于整个行程，从而支持直接使用总距离除以平均速度。
- 若误解“平均速度”可能随路段变化，学生可能误选 D（“无法判断”）。因此，对现实情境的理解有助于避免常见误区。
- 对速度表功能的常识性理解，有助于正确建模。

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5. 能力 #8：清晰表述解法，展示和/或总结并呈现中间数学结果

（在当前题型下相关性较弱）

- 本题为选择题，不要求书面解释。
- 但在教学场景中，若要求学生说明为何 D 不正确（即：平均速度确实足以计算总时间），则此项能力会被激活。